Modeling CIN Cancer Evolution: A Dual Population Approach for Astrocytoma and Adenocarcinomas

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Abstract:

Different cancer types show a high degree of variance in chromosome count distribution among their populations. One identifiable trend across multiple cancer source tissues is chromosome count bimodality. This is partially explained by whole genome doubling (WGD). However, chromosomally unstable cancers such as those exhibiting bimodality do not share a common chromosome distribution, suggesting alternate mechanisms at work. This discrepancy may be explained by evolutionary mechanisms, where both the bimodal distribution and its change between tissue types represent multiple viable evolutionary strategies within a single cancer type, with the viability of those strategies changing across tissues. This would explain the observed discrepancy as multiple cancer populations exploiting different evolutionary and/or environmental strategies within a tissue.

To better explore this explanation, this project models tumor evolution as a Markov chain representing chromosome missegregation over time, comparing that model to real world data, and performing a grid search to best fit the model to said data. Simulated astrocytoma cells were shown to exhibit radically different selective pressures after WGD, wherein an alternate high variance strategy incentivising chromosome loss best fit observations. This suggests at least two evolutionarily distinct populations within the astrocytoma type, a low chromosome-variance diploid population and a smaller high chromosome-variance polyploid population.

The model was further applied to pancreatic and prostrate adenocarcinomas. Simulated pancreatic adenocarcinomas exhibited a generalized high variance strategy, with a larger proportion of the observed population spread across elevated chromosome counts, WGD cells showing increased rates of aneuploidy. Simulated prostrate adenocarcinomas, by contrast, exhibited a generalized low variance strategy with a significantly elevated chance of WGD, wherein tetraploid cells do not exhibit increased chromosome missegregation rate after WGD.

Overall, this model suggests cancer cells exhibit multiple viable chromosomal evolutionary strategies within a single cancer type, best characterized as distinct populations.

Acknowledgement:

This work was prompted and inspired by the unpublished notes of Kelly Brock, reviewed and catalogued as part of the work component of this course's summer project. Kelly Brock attempted to model astrocytoma chromosome missegregation in 2016 by incorporating a whole genome doubling parameter. She was ultimately unsuccessful, but identified the bimodal characteristic of cancer chromosomal counts such as astrocytoma, demonstrated a hypothetical bimodal model could overlap the data, and showed that a simple model of chromosome missegregation with whole genome doubling was unable to accurately capture both actual counts and a bimodal data distribution.

This project has made use of some sources she referenced, most notably the Mitelman Database for chromosome counts. This project does not use any code written by Kelly Brock, and none was present in the notes recovered. This project does independently recreate one graph she created using more modern data: a comparison of astrocytoma chromosome types across tumor morphologies. All other work in the project was independent.

Multiple attempts have been made to contact Kelly Brock, which have not yet received a reply. A copy of her notes most relevant to this project are attached.

Introduction:

Chromosomal Instability (CIN) in a cell is characterized by genomic instability resulting in alterations to chromosome count and structure. CIN may be prompted by a wide variety of genetic processes, many of them involved with mitosis and aberrations therein. Notably, CIN plays a role in tumor cells, whereby cancer populations exhibiting aneuploidy may show significantly divergent karyotypes from the original organism. (A'Hern et al., 2013)

Cancer populations with different karyotypes from such aneuploidy have different degrees of fitness, and thus expected pathologies. While eukaryotes as a whole organism are usually diploid, changes to individual cell ploidy as part of an organism's development cycle or specific tissue function are relatively common. Similarly, cancer populations with different karyotypes from aneuploidy have varying levels fitness (and thus expected pathologies), specific to the cell type. (Storchova & Kuffer, 2008)

These chromosomal alterations in CIN cells represent an evolutionary process, altering genetic variability of cancer cells. However, such changes are difficult to track in vivo. Multiple karyotypes may exist within same organism (or even the same tumor). Only recently has single cell sequencing made genotyping spatial sub-populations within tumors practical, if not necessarily widely accessible. (A'Hern et al., 2013) (Haughey et al., 2023)

Still, wider trends in CIN cell behavior can be observed in aggregate data. For example, a trend toward chromosome number bimodality is common across multiple tumor types, wherein CIN cell chromosome counts are concentrated around specific separated peaks. One mechanism by which CIN cells alter chromosome number is whole genome doubling (WGD), wherein cell nuclei fail to separate during attempted mitosis, resulting in a single tetraploid cell. This mechanism would account for the bimodal portion of some distributions, and a plausible pathway to higher chromosome numbers. (Vittoria et al., 2023)

Observed CIN tumor bimodal karyotypes, however, often do not center around the tetraploid chromosome numbers expected from simple WGD, and instead show a different distribution pattern around each peak, differing between cell types. Better characterization of polyploidy tumor behavior, such as modeling how their karyotypes change over time, has many implications for wider cancer pathology. If this chromosome discrepancy between represents a range of differing viability based on tumor source tissue and environment, then multiple populations of cancer cells are functionally exploiting different evolutionary strategies, even within their nominal cancer type. This would manifest in observable population trends, such as differing chromosome count distributions among cancer cell populations based on their tissue type. (Storchova & Kuffer, 2008) (Ghosh et al., 2024)

A dual population explanation for bimodal discrepancies can be explored through evolutionary modelling. As CIN is an evolutionary process, evolutionary models can be applied and bioinformatically tested against real world data. For example, while each tumor arises spontaneously, they display similar aggregate trends that can be tracked together as if they were a population with shared traits. If tumors that have undergone WGD can be characterized by a discrete population with different missegregation rates than diploid tumors, a model tracking chromosome changes over time can demonstrate this. (Somarelli & Johnson, 2024)

While there has been recent renewed interest in evolutionary oncology, few studies have applied dual population evolutionary models to existing cancer data. This project attempts to demonstrate one method by which such models can be implemented, fitting a Markov Chain model of chromosome missegregation to real world astrocytoma, pancreatic adenocarcinoma, and prostrate adenocarcinoma data via a grid search. (Haughey et al., 2023)

By modeling evolutionary chromosome missegregation in this manner, this project showcases how different evolutionary strategies and pressures can affect tumor cells within a given cancer type, resulting in multiple discrete, identifiable populations. By exploring the functional implications of simulated selective pressures, the project aims to demonstrate how these populations may change evolutionary strategies dependent on their source tissue, resulting in different chromosomal effects beyond increased aneuploidy tolerance from the single mechanism of WGD.

Methods:

Scripting was performed in R 4.5.1, with the dplyr 1.2.1, tidyr 1.3.2, stringr 1.6.0, ggplot2 4.0.3, forreach 1.5.2, and doparallel 1.0.17 packages. R was used due to its broad support of statistical tools and analysis software, on top of prior familiarity. dplyr was used for statistical analysis. tidyr was used to facilitate piping and make scripting more legible. stringr was used for text extraction of chromosome numbers from database files. ggplot2 was used for plot creation and data display. forreach and doparallel were used to allow multithreading in the grid search. (R Core Team, R, 2025), (Whickam et al., dplyr, 2026), (Whickham et al.,tidyr, 2026), (Whikham, stringr, 2025), (Whicham, ggplot2, 2016), (Daniel et al, forreach, 2022), (Daniel et al, doparallel, 2022)

Github Link: https://github.com/WFeinman/BINF6999-Summer-Project---Modeling-CIN-Cancer-Evolution

The broad methodology of this project can be broken down into 3 steps:

Step 1 - Find: Take candidate cancer, confirm presence of bimodal distribution in modern population data.

Step 2 - Model: Set up a Markov Chain model, mathematically outlining possible changes to chromosome number over time. (Øksendal, 2005)

Step 3 - Refine: Perform a Maximum Likelihood Estimate (MLE) via log-likelihood (LL) grid search to determine missegregation parameters for both groups in dataset, compare populations. (Dekking et al., 2005)

Step 1 - Find:

The “Mitelman Database of Chromosome Aberrations and Gene Fusions in Cancer” is an ongoing catalogue of gene + chromosome changes in sampled cancers, organized via ISCN notation and formatting. It includes source tissue, cancer type, and chromosome count, downloadable as a .csv file via their "Cases Cytogenetics Searcher" tool. This data can be processed to overview candidate cancer types for bimodality. (Mitelman, 2026) (Wang & LaFramboise, 2019)

For quality control, NA entries were trimmed, cancers with chromosome ranges recorded were averaged and rounded down. The cancer type selected for modeling were Grade

I-IV astrocytomas, due to high representation in the database (627 samples listed), a bimodal chromosome distribution with clearly differentiated distribution patterns around each peak, and prior modeling work performed (see Acknowledgement). Additionally, the grading system for astrocytoma prognosis classification, is recorded in the database, lending itself to further analysis. (Gerges et al., 2013)

Other cancer types (pancreas and prostrate adenocarcinomas) were also examined as case studies in alternate chromosome distribution patterns.

Step 2 – Model:

A Markov Chain model to mathematically describe missegregation over time. To do so, tumor cells are simulated via population fraction distributed across a range of chromosome counts, from 0 to 150. The model outlines 2 groups, Group A, and Group B. Group A represents cells which remain roughly diploid. Group B represents cells which have undergone WGD, becoming roughly tetraploid.

First, a transition matrix is generated for Group A cells, representing the probability of simulated cells gaining or losing a chromosome each cycle, calculated per chromosome. Thus, cells with a larger chromosome number have a greater chance of aneuploidy occurring per cycle, though missegregation has an equal chance of causing either a chromosome gain or a chromosome loss. As increased ploidy has been linked to an increase in chromosomal instability, an additional transition matrix with a separate probability distribution is generated for Group B cells. (Ghosh et al., 2024)

The simulated tumor cell population starts with the entire the population at 46 chromosomes in Group A. Each cycle (t), all cells have a chance to gain/lose chromosomes via the missegregation rate. Additionally, Group A cells have a chance to undergo WGD, doubling their chromosome count and transitioning to Group B.

After missegregation, fitness parameters are applied to all cells, wherein a chromosome number is specified alongside a selection chance for cells to be removed in any given time step. The greater the difference between a given cell's chromosome number and the fitness parameter, the more likely that cell is to be removed. This number is divided by a directional selection strength variable,

The net selection function for chromosome gain in Group A above the fitness parameter is specified as follows:

For (x_vals > peakA): FA <- exp(-0.5 * ((x_vals - peakA) / sigmaA_gain)^2)

Where x_vals is the simulated cells current chromosome count, peakA is the fitness parameter representing optimal chromosome number for Group A, and sigma_gain is the selection strength acting upon cells with chromosome numbers above peakA. A similar function characterizes selection below the fitness peak:

Else: FA <- exp(-0.5 * ((x_vals - peakA) / sigmaA_loss)^2))

Group B cells have a separately tracked missegregation rate, fitness parameter, and bi-directional selection chance than Group A cells. By doing so, the model can simulate different chromosome missegregation and selection strategies for both populations, and allow independent selection weights to be applied to chromosome gain and chromosome loss.

Step 3 - Refine:

A Maximum Likelihood Estimate (MLE) is a statistical technique to fit models by comparing their relative likelihood for the actual data to have occurred under a given model. The more probable it is for the actual data to occur under the model, the more positive its MLE.

Log-likelihood (LL) is a specific form of this technique which compares the sum of the logarithm of each probability in the comparison, rather than taking the product. Summation is less computationally costly (after conversion) than multiplication, which is useful when performing hundreds of thousands to millions of computational comparisons. It is also somewhat more resilient to disruption by outliers, as logarithmic summation means a single very-low-chance outlier in the actual dataset has less of an impact on which model is selected.

For assurance of fit and to facilitate reproducibility, a grid search is performed to test a range of values for each mathematical parameter specified in the Markov Chain model. Namely, WGD chance, missegregation rate, fitness parameter, and selection chance for both Group A and Group B. To do so, the model is simulated for each possible parameter combination in a specified range, determining which combination produces the best fitting model by MLE LL.

It should be noted that grid searches grow exponentially more computationally intensive with each added parameter and testing range increase. A full exhaustive search of each variable here would require either a computing cluster or some weeks of single-computer computation, even with multithreading implemented. As such, the models attached were written to be a demonstration of the modeling workflow, rather than a definitive characterization. The astrocytoma model should run on a single computer in roughly an hour. The pancreatic and prostrate adenocarcinoma models should run on a single computer in about twenty minutes.

Results:

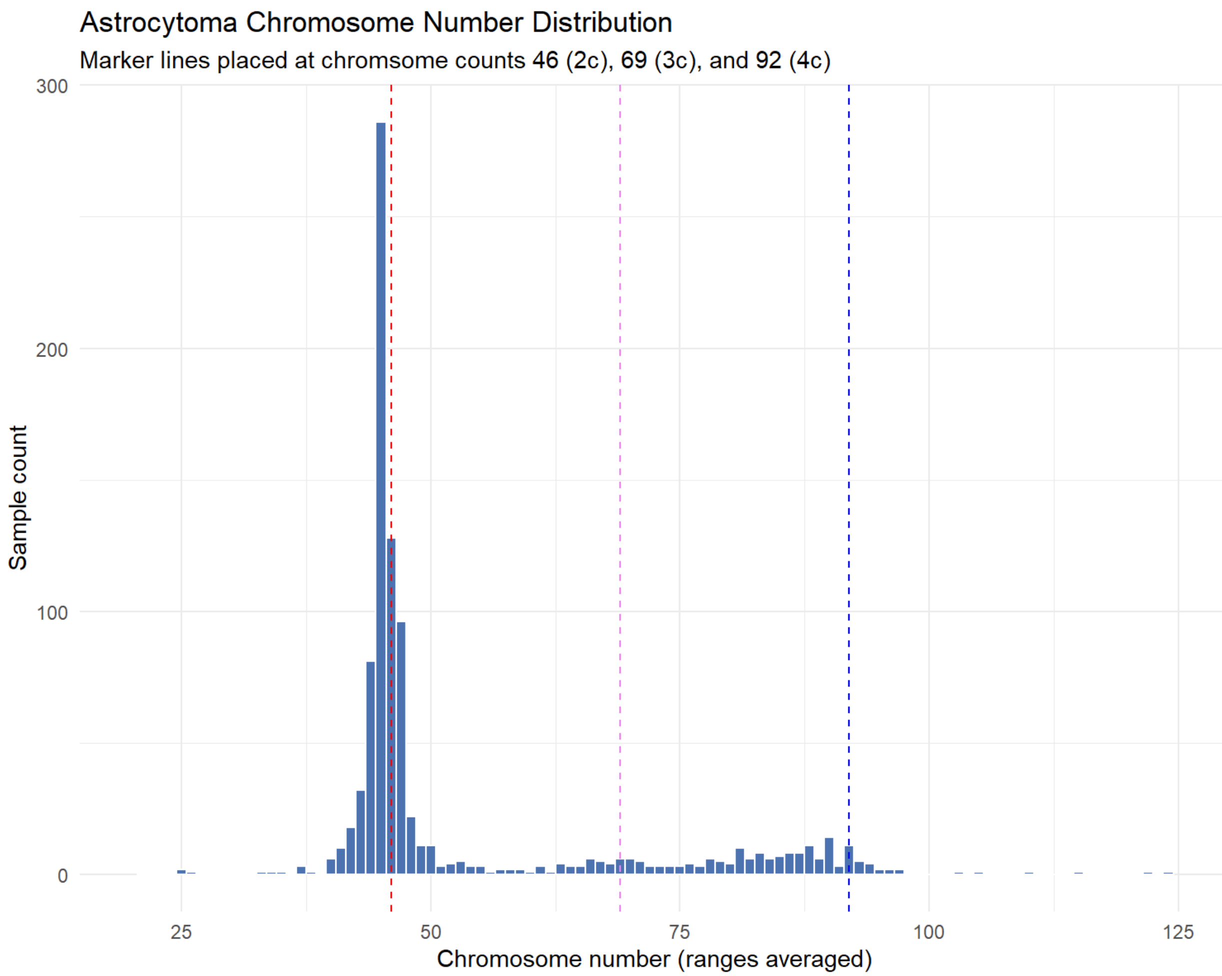

Figure 1: Distribution of chromosome numbers among Astrocytoma (Grade I-IV) samples in the Mitelman Database. Note the bimodal distribution pattern, and how it varies around each relative peak. Solitary samples with chromosome numbers above 125 (including one observed sample of 186) were excluded from this graph for legibility purposes.

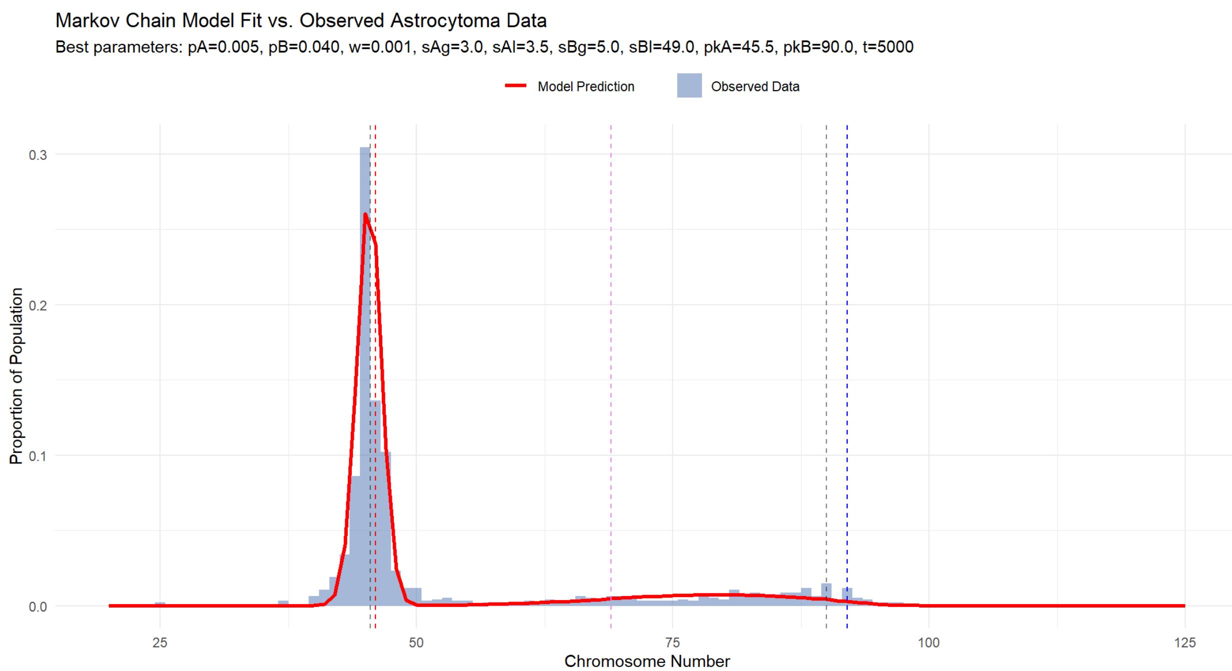

Figure 2: Markov Chain model outlines observed data, with some irregularities. Note that both missegregation chance (pA/B) and selection strength (sA/Bg/l) vary between Group A and Group B, representing difference in chromosome stability and tolerance of aneuploidy. Also note that Group B has very different values for its selection strength parameters, representing more strict selection against chromosome gain above the fitness parameter (pkB) than against chromosome loss below it.

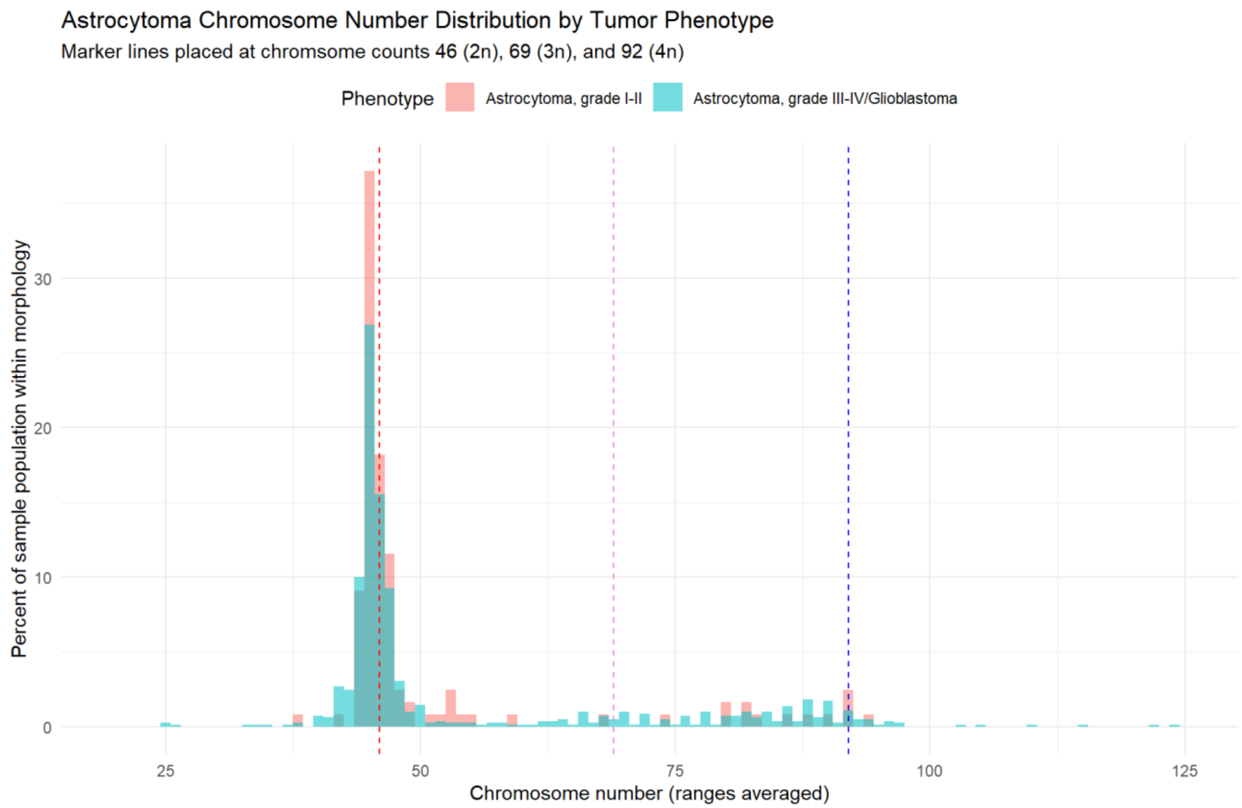

Figure 3: Splitting Astrocytomas by progression stage: Increased chromosome number displays some correlation with malignant phenotypes vs benign tumors.

Morphology	Population (n)	n=>55	n=>75
Astrocytoma, grade I-II	121	16	12
Astrocytoma, grade III-IV/Glioblastoma	818	197	135

Table 1: Comparison of Astrocytoma malignancy status against elevated chromosome count. Derivative statistics below.

Fraction of total Astrocytoma (grade I-II) samples over 55 chromosomes: 0.132

Fraction of total Astrocytoma (grade III-IV/Glioblastoma) samples over 55 chromosomes: 0.240

Relative fraction of malign samples with 55+ chromosomes over benign samples with 55+ chromosomes: 1.821

Fraction of total Astrocytoma (grade I-II) samples over 75 chromosomes: 0.099

Fraction of total Astrocytoma (grade III-IV/Glioblastoma) samples over 75 chromosomes: 0.165

Relative fraction of malign samples with 75+ chromosomes over benign samples with 75+ chromosomes: 1.664

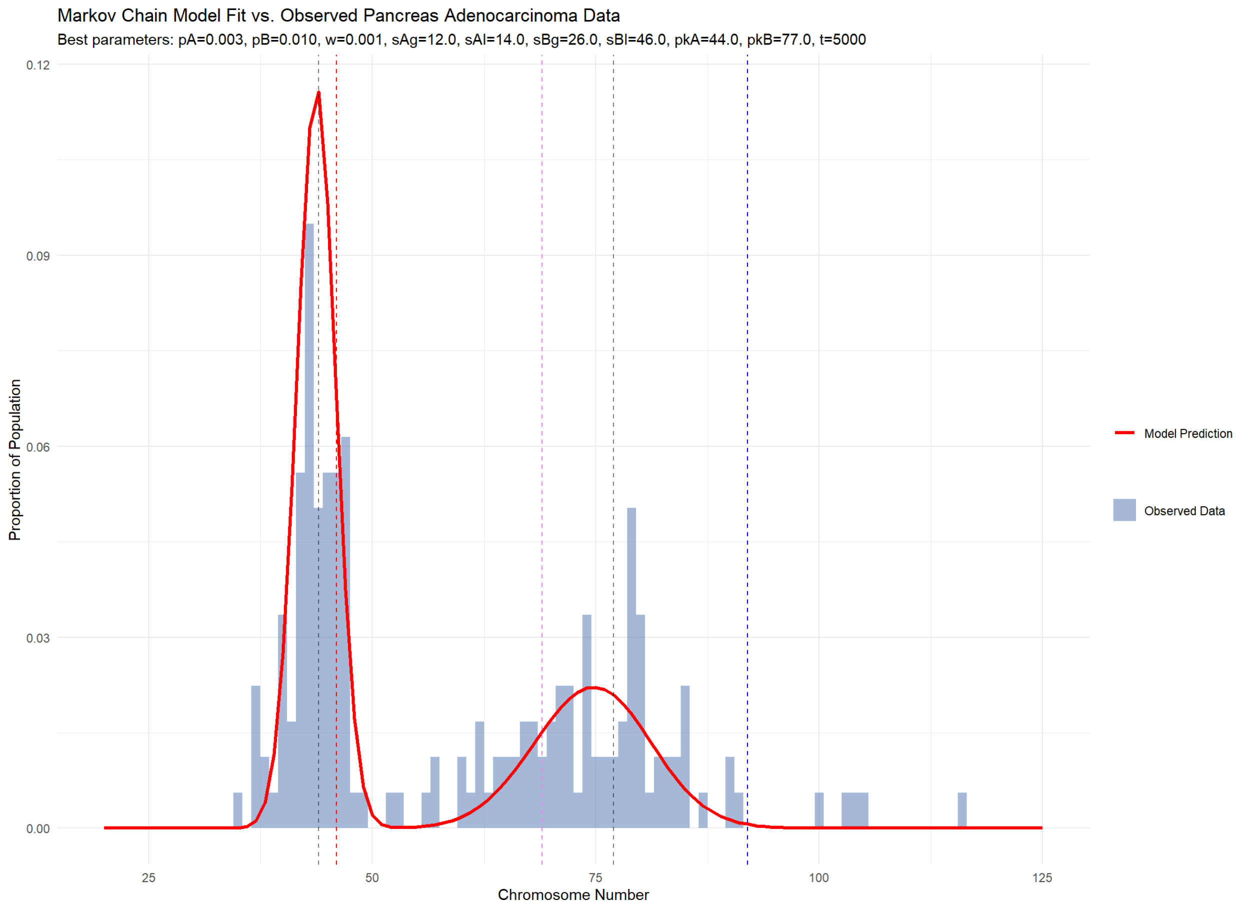

Figure 4: Markov model tracks observed pancreatic adenocarcinoma data from the Mitelman database. While calculated missegregation rates was lower than the Astrocytoma model, the model indicates a much greater tolerance for gain-of-chromosome in both Group A and Group B cells.

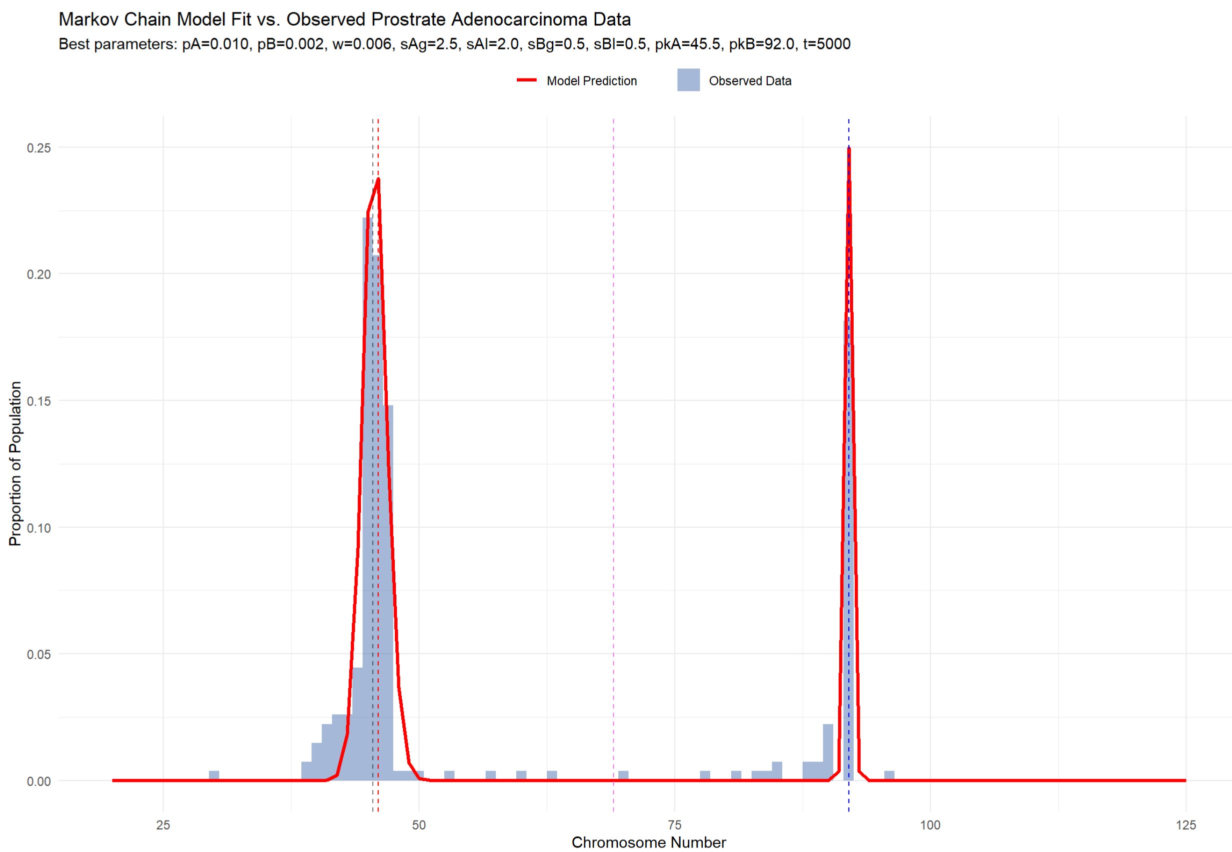

Figure 5: Markov model tracks observed pancreatic adenocarcinoma data from the Mitelman database. Unlike previous models, Group B does not display an increased tolerance for aneuploidy, despite a much higher simulated rate of WGD.

Discussion:

The Markov model suggests differing evolutionary parameters exist within astrocytomas, depending on WGD status. The wide chromosome distribution of the simulated Group B cells was facilitated by a high missegration rate. Prior literature suggests this may play into an evolutionary strategy, wherein polyploid cells have a wider number of mechanisms for genetic variation, albeit at increased metabolic cost and risk of apoptosis (Ghosh et al., 2024).

By contrast, simulated Group A cells which have not undergone WGD tended to towards a lower missegration rate. Diploid tumors, compared to polyploidy counterparts, tend to be more chromosomally stable and less tolerant of chromosome number variation.

While a correlation is suggested between Astrocytoma chromosome count over 55 and malignant phenotypes, a caveat should be noted that this database is an aggregation of karyotypes and genetic investigation from focused investigative publishings, only a portion of which are population studies. As such, malignant phenotypes are more likely to be represented due to sampling bias, simply because malignant cancers are more likely present identifiable symptoms that allow them to be found and subsequently studied.

It should be noted, that these chromosome numbers are approximations by necessity. CIN lines often have massive alterations to chromosomal content due to large scale rearrangements such as chromosome fragmentation and fusion, obscuring just how much and what parts of specific chromosomes have been duplicated or discarded without sequencing data. Chromosome ranges are specified in the Mitelman dataset, for example, because the question of whether a chromosome that has split in half should be counted as one or two may make the difference between a count of 45 chromosomes or a count of 136 chromosomes. (Krupina et al., 2024) (Ransom et al., 1992)

Similarly, it should be noted that chromosomal content itself is liable to shift from such rearrangements. What appears to be a Chromosome 1 may in fact be a composite of 1, 2, and 5. This, in part, is why this model aimed for an abstract fitness parameter keyed off chromosome number. The actual content of the chromosomes themselves can be highly variable. (Krupina et al., 2024)

One of the shortcomings of the model was an estimation overshoot around the 75c mark. A number of potential explanations for this exist.

1. Outside Model Range: This disappear with a higher fidelity gird search, albeit computationally expensive to perform.
2. Sample Bias: Cells of that number may exist in higher proportions than data exists for, they just haven't been sampled.

3. Trimodal Population: The discrepancy could hypothetically be explained by a third population with different characteristics, or an unknown mechanism to jump to another copy number.
4. Unknown Selection Interaction: Some form of disruptive selection may exist around that chromosome number, such as an evolutionary bottleneck. Research in colorectal cancer suggests evolutionary bottlenecks on copy number in CIN lines may act as stabilizing influence on karyotype, despite cell instability. If a similar phenomenon is occurring here, that would go some way to explaining this behavior, and similar patterns may be found by modeling other cancers. (Cross et al., 2026)

Distinguishing populations may clarify prognosis. By better characterizing how populations arise and evolve within cancer types, a clearer diagnostic picture is presented. Identifying higher risk tumor populations may facilitate early intervention. Similarly, knowing the evolutionary variability of a tumor population can inform treatment choice and expected outcome. (Gerges et al., 2013)

On a wider scale, distribution patterns were shown differ by tissue type. In pancreatic adenocarcinomas, elevated chromosome counts are more common, as is aneuploidy among near-diploid tumors. The model attributes this to a high selection tolerance for chromosome gain in both diploid and polyploidy cells.

However, not all bimodal populations show different distribution patterns in polyploidy variants. Prostrate adenocarcinoma bimodality is far more dominated by WGD chance than missegregation. Unlike the other examined cancer types, tetraploid cells do not exhibit an increased rate of missegregation relative to diploid cells within the dataset. This suggests the increase in genomic instability following polyploidy may not be a universal trend in cancers. (Ghosh et al., 2024)

In conclusion, chromosome count discrepancies in real world tumor karyotype data can be modeled by the existence of multiple, evolutionarily distinct populations within a given cancer type, each exploiting different chromosome missegregation strategies. Astrocytoma modeling suggests a change in strategy from low variance to high variance can be prompted by WGD. Pancreas adenocarcinoma modeling shows that while high variance can be increased by WGD, a high variance strategy still remains viable without it. The prostrate adenocarcinoma model shows that WGD does not necessitate an increase in chromosome missegregation, with a low variance strategy remaining viable at elevated chromosome counts. These changes in chromosome strategy vary both across and within tissue types. This suggests the existence of a wide range of viable evolutionary strategies for cancer across the organism, exploited by multiple tumor populations.

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